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Pixel array including time-of-flight sensors

Abstract

A pixel array may include a group of time-of-flight (ToF) sensors. The pixel array may include an image sensor comprising a group of pixel sensors. The image sensor may be arranged among the group of ToF sensors such that the image sensor is adjacent to each ToF sensor in the group of ToF sensors.

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Background/Summary

RELATED APPLICATION (1) This application is a division of U.S. patent application Ser. No. 17/249,969, filed Mar. 19, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

(1) Complementary metal oxide semiconductor (CMOS) image sensors utilize light-sensitive CMOS circuitry, referred to as pixel sensors, to convert light energy into electrical energy. A pixel sensor typically includes a photodiode formed in a silicon substrate. As the photodiode is exposed to light, an electrical charge is induced in the photodiode. The photodiode may be coupled to a switching transistor, which is used to sample the charge of the photodiode. Colors may be determined by placing color filters over photodiodes of a CMOS image sensor.

(2) Time-of-Flight (ToF) sensors (e.g., sensors that use germanium-on-silicon (GeSi) technology to enable ToF sensing) can be used in a system designed to detect distances to objects in an area. Generally, a given ToF sensor detects a phase difference between a signal transmitted by the system and a corresponding signal received by the given ToF sensor (after reflection of the signal by an object in the area). This phase difference can be used to determine the distance to the object that reflected the signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a diagram of an example environment in which systems and/or methods described herein may be implemented.
- (3) FIGS. 2A-2D are diagrams associated with example pixel arrays described herein.
- (4) FIGS. 3A-3C are diagrams associated with example pixel arrays described herein.
- (5) FIG. 4 is a diagram of an example pixel array described herein.
- (6) FIGS. 5A-5L are diagrams of an example implementation described herein.
- (7) FIGS. 6A and 6B are diagrams of an example pixel array described herein.
- (8) FIG. 7 is a diagram of absorption data described herein.
- (9) FIG. 8 is a diagram of example components of one or more devices of FIG. 1.
- (10) FIG. 9 is a flowchart of an example process relating to forming a pixel array.

DETAILED DESCRIPTION

(11) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(12) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(13) In some cases, outputs from an array of ToF sensors can be used to generate a distance image (also referred to as a ToF image or a depth image) that indicates distances to objects in an area. However, while the distance image indicates distances to the objects in the area, the distance image does not provide a color image of the area. Further, outputs from an array of image sensors (e.g., an array of CMOS image sensors) can be used to generate a color image that indicates colors of objects in an area. However, while the color image indicates colors of the objects in the area, the color image does not provide a distance image of the area.

(14) Some implementations described herein describe a pixel array that includes a plurality of ToF sensors and a plurality of image sensors (e.g., a plurality of CMOS image sensors, such as a plurality of red-green-blue (RGB) image sensors). In some implementations, outputs of the ToF sensors and outputs of the image sensors may be used to generate an image that indicates both distance to and color of objects in an area (herein referred to as a three-dimensional (3D) ToF color image). That is, the pixel array described herein enables distance information determined by ToF sensors and color information determined by image sensors to be combined to enable generation of a 3D ToF color image that indicates both distances to and colors of objects in an area.

(15) A 3D ToF color image may be useful in, for example, 3D module construction, which can be used in a variety of applications, such as a medical application, a virtual reality (VR) application, an augmented reality (AR) application, a 3D printing application, or an autonomous vehicle application, among other examples.

(16) FIG. 1 is a diagram of an example environment **100** in which systems and/or methods described herein may be implemented. As shown in FIG. 1, environment **100** may include a plurality of semiconductor processing tools **102-112** and a wafer/die transport tool **114**. The plurality of semiconductor processing tools **102-112** may include a deposition tool **102**, an exposure tool **104**, a developer tool **106**, an etch tool **108**, a planarization tool **110**, an ion implantation tool **112**, and/or another type of semiconductor processing tool. The tools included in example environment **100** may be included in a semiconductor clean room, a semiconductor foundry, a semiconductor processing facility, and/or manufacturing facility, among other examples.

(17) The deposition tool **102** is a semiconductor processing tool that includes a semiconductor processing chamber and one or more devices capable of depositing various types of materials onto a substrate. In some implementations, the deposition tool **102** includes a spin coating tool that is capable of depositing a photoresist layer on a substrate such as a wafer. In some implementations, the deposition tool **102** includes a chemical vapor deposition (CVD) tool such as a plasma-enhanced CVD (PECVD) tool, a high-density plasma CVD (HDP-CVD) tool, a sub-atmospheric CVD (SACVD) tool, an atomic layer deposition (ALD) tool, a plasma-enhanced atomic layer deposition (PEALD) tool, or another type of CVD tool. In some implementations, the deposition tool **102** includes a physical vapor deposition (PVD) tool, such as a sputtering tool or another type of PVD tool. In some implementations, the example environment **100** includes a plurality of types of deposition tools **102**.

(18) The exposure tool **104** is a semiconductor processing tool that is capable of exposing a photoresist layer to a radiation source, such as an ultraviolet light (UV) source (e.g., a deep UV light source, an extreme UV light (EUV) source, and/or the like), an x-ray source, an electron beam (e-beam) source, and/or the like. The exposure tool **104** may expose a photoresist layer to the radiation source to transfer a pattern from a photomask to the photoresist layer. The pattern may include one or more semiconductor device layer patterns for forming one or more semiconductor devices, may include a pattern for forming one or more structures of a semiconductor device, may include a pattern for etching various portions of a semiconductor device, and/or the like. In some implementations, the exposure tool **104** includes a scanner, a stepper, or a similar type of exposure tool.

(19) The developer tool **106** is a semiconductor processing tool that is capable of developing a photoresist layer that has been exposed to a radiation source to develop a pattern transferred to the photoresist layer from the exposure tool **104**. In some implementations, the developer tool **106** develops a pattern by removing unexposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by removing exposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by dissolving exposed or unexposed portions of a photoresist layer through the use of a chemical developer.

(20) The etch tool **108** is a semiconductor processing tool that is capable of etching various types of materials of a substrate, wafer, or semiconductor device. For example, the etch tool **108** may include a wet etch tool, a dry etch tool, and/or the like. In some implementations, the etch tool **108** includes a chamber that is filled with an etchant, and the substrate is placed in the chamber for a particular time period to remove particular amounts of one or more portions of the substrate. In some implementations, the etch tool **108** may etch one or more portions of the substrate using a plasma etch or a plasma-assisted etch, which may involve using an ionized gas to isotopically or directionally etch the one or more portions.

(21) The planarization tool **110** is a semiconductor processing tool that is capable of polishing or planarizing various layers of a wafer or semiconductor device. For example, a planarization tool

110 may include a chemical mechanical planarization (CMP) tool and/or another type of planarization tool that polishes or planarizes a layer or surface of deposited or plated material. The planarization tool **110** may polish or planarize a surface of a semiconductor device with a combination of chemical and mechanical forces (e.g., chemical etching and free abrasive polishing). The planarization tool **110** may utilize an abrasive and corrosive chemical slurry in conjunction with a polishing pad and retaining ring (e.g., typically of a greater diameter than the semiconductor device). The polishing pad and the semiconductor device may be pressed together by a dynamic polishing head and held in place by the retaining ring. The dynamic polishing head may rotate with different axes of rotation to remove material and even out any irregular topography of the semiconductor device, making the semiconductor device flat or planar.

(22) The ion implantation tool **112** is a semiconductor processing tool that is capable of implanting ions into a substrate. The ion implantation tool **112** may generate ions in an arc chamber from a source material such as a gas or a solid. The source material may be provided into the arc chamber, and an arc voltage is discharged between a cathode and an electrode to produce a plasma containing ions of the source material. One or more extraction electrodes may be used to extract the ions from the plasma in the arc chamber and accelerate the ions to form an ion beam. The ion beam may be directed toward the substrate such that the ions are implanted below the surface of the substrate.

(23) Wafer/die transport tool **114** includes a mobile robot, a robot arm, a tram or rail car, an overhead hoist transport (OHT) system, an automated materially handling system (AMHS), and/or another type of device that is used to transport wafers and/or dies between semiconductor processing tools **102-112** and/or to and from other locations such as a wafer rack, a storage room, and/or the like. In some implementations, wafer/die transport tool **114** may be a programmed device that is configured to travel a particular path and/or may operate semi-autonomously or autonomously.

(24) The number and arrangement of devices shown in FIG. **1** are provided as one or more examples. In practice, there may be additional devices, fewer devices, different devices, or differently arranged devices than those shown in FIG. **1**. Furthermore, two or more devices shown in FIG. **1** may be implemented within a single device, or a single device shown in FIG. **1** may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) of environment **100** may perform one or more functions described as being performed by another set of devices of environment **100**.

(25) FIGS. **2A-2D** are diagrams associated with an example pixel array **200** described herein. In some implementations, the pixel array **200** may be included in an imaging system configured to generate 3D ToF color images. FIG. **2A** is a top view of a first example layout of the pixel array **200**, and FIGS. **2B** and **2C** are cross-sectional views along the lines labeled **1** and **2** in FIG. **2A**, respectively. As shown in FIGS. **2A**, the pixel array **200** may include a group of ToF sensors **202** (e.g., ToF sensors **202a** through **202d**) and an image sensor **204** comprising a group of pixel sensors **206** (e.g., pixel sensors **206a** through **206d**).

(26) The ToF sensor **202** is a component capable of providing ToF sensing to determine distance information for a signal reflected to ToF sensor **202** (e.g., a signal transmitted by a transmission device associated with the pixel array **200** that is reflected back to the pixel array **200**). In some implementations, the distance information determined by ToF sensor **202** indicates a distance to objects in an area in an environment of the ToF sensor **202**. In some implementations, the distance information is determined by detecting a phase difference between the transmitted signal and the corresponding signal received by the ToF sensor **202** (after reflection of the signal by an object in the area). This phase difference can be used to determine the distance to the object that reflected the signal. In some implementations, the ToF sensor **202** may utilize germanium-on-silicon (GeSi) technology to enable ToF sensing. In some implementations, a GeSi ToF sensor **202** provides high quantum efficiency and a high demodulation contrast at a high operation frequency, thereby

enabling improved depth accuracy when determining distance information. In the example of FIG. 2A, a size $d_{\text{sub.t1}}$ of the ToF sensor **202** may be, for example, approximately 10 microns (μm) and a size $d_{\text{sub.t2}}$ of the ToF sensor **202** may be, for example, approximately 6 μm . However, other sizes $d_{\text{sub.t1}}$ and $d_{\text{sub.t2}}$ of the ToF sensor **202** are within the scope of the present disclosure. In some implementations, a size of the ToF sensor **202** may be represented as a ratio of a length (e.g., $d_{\text{sub.t1}}$) to a width (e.g., $d_{\text{sub.t2}}$) of the ToF sensor **202**, or length/width (e.g., $d_{\text{sub.t1}}/d_{\text{sub.t2}}$), such as 1.667. However, other ratios of the length to the width of the ToF sensor **202** are within the scope of the present disclosure.

(27) The image sensor **204** is a component including a group of pixel sensors **206** to determine color information for incident light at the pixel array **200**. In some implementations, image sensor **204** may be a CMOS image sensor. In some implementations, the group of pixel sensors **206** may include one or more red pixel sensors, one or more green pixel sensors, one or more blue pixel sensors, one or more yellow pixel sensors, one or more white pixel sensors, and/or one or more other types of pixel sensors configured to sense incident light in the visible light wavelength range. For example, in some implementations, the pixel sensors **206a** and **206c** may be green pixel sensors, the pixel sensor **206b** may be a blue pixel sensor, and the pixel sensor **206d** may be a red pixel sensor (e.g., such that the image sensor **204** is an RGGB sensor). In some implementations, a given pixel sensor **206** may be formed and/or configured to sense a wavelength of incident light associated with a particular color of visible light. For example, a red light pixel sensor may be a visible light pixel sensor that is formed and/or configured to sense a wavelength range of incident light corresponding to a red component of visible light (e.g., to provide red color information for the incident light), a green light pixel sensor may be a visible light pixel sensor that is formed and/or configured to sense a wavelength range of incident light corresponding to a green component of visible light (e.g., to provide green color information for the incident light), and a blue light pixel sensor may be a visible light pixel sensor that is formed and/or configured to sense a wavelength range of incident light corresponding to a blue component of visible light (e.g., to provide blue color information for the incident light). In some implementations, the pixel sensors **206** of the image sensor **204** in the pixel array **200** may be used to sense and obtain color information (e.g., color saturation information, color intensity information, color distribution information, and/or another type of color information) for incident light at the pixel array **200**. In some implementations, sizes $d_{\text{sub.p1}}$ and $d_{\text{sub.p2}}$ of a given pixel sensor **206** may be, for example, in a range from approximately 1 μm to approximately 2.5 μm . However, other sizes $d_{\text{sub.p1}}$ and $d_{\text{sub.p2}}$ of a given pixel sensor **206** are within the scope of the present disclosure.

(28) As shown in FIGS. 2A, in some implementations, the image sensor **204** is arranged among the group of ToF sensors **202** such that the image sensor **204** is adjacent to each ToF sensor **202** in the group of ToF sensors **202**. That is, the image sensor **204** may be arranged between the group of ToF sensors **202** in the pixel array **200**. In some implementations, the image sensor **204** is separated from each ToF sensor **202** of the group of ToF sensors **202** by an isolation region **208**. Similarly, in some implementations, each ToF sensor **202** of the group of ToF sensors **202** is separated from other ToF sensors **202** in the group of ToF sensors **202** by an isolation region **208**. In some implementations, the isolation region **208** serves to reduce or prevent cross-talk between a given pair of sensors (e.g., between a pair of ToF sensors **202** or between a ToF sensor **202** and the image sensor **204**), thereby improving performance of the pixel array **200** (e.g., in terms of saturation, accuracy, noise, contrast, brightness, hue and saturation, light sensitivity, or contour sharpness). In some implementations, a size d_i of the isolation region **208** may be approximately 1 μm . In some implementations, the isolation region **208** having a size of approximately 1 μm provides adequate isolation without significantly impacting an overall size of the pixel array **200**. However, other sizes d_i of the isolation region **208** are within the scope of the present disclosure. Notably, isolation performance by the isolation region **208** versus an overall area of the pixel array **200** is a trade-off, meaning that isolation performance within the pixel array **200** can be balanced against an overall

area of the pixel array **200**, as desired. In general, the compact nature of the pixel array **200** reduces and/or minimizes unused gaps or portions between the ToF sensor **202** and the image sensor **204** in the pixel array **200**, which increases the sensor density and increases spatial utilization.

(29) In some implementations, the ToF sensors **202** surround the image sensor **204** such that the image sensor **204** is centered in the ToF sensors **202**. For example, as shown in the first example layout of FIG. 2A, the group of ToF sensors **202** may surround the image sensor **204** such that the pixel array **200** forms a rectangular pattern (e.g., a square pattern). In such a case, to enable the formation of the rectangular pattern, a first set of ToF sensors **202** in the group of ToF sensors **202** may be oriented in a first direction and a second set of ToF sensors **202** in the group of ToF sensors **202** may be oriented in a second direction that is different from the first direction. Taking the first example layout shown in FIG. 2A as an example, ToF sensor **202a** and ToF sensor **202c** are oriented with respective lengths along a first direction (e.g., a horizontal direction in FIG. 2A), while ToF sensor **202b** and ToF sensor **202d** are oriented with respective lengths along a second direction (e.g., a vertical direction in FIG. 2A). In this example, ToF sensors **202a** and **202c** are oriented perpendicular relative to ToF sensors **202b** and **202d**. In some implementations, a rectangular pattern enabled by the differently oriented sets of ToF sensors **202** improves spatial utilization of a semiconductor die that includes pixel array **200**. For example, the rectangular pattern of the first example layout enables multiple pixel arrays **200** (and/or multiple portions of pixel arrays **200** having the first example layout) to be arranged adjacent to one another to form a larger pixel array (e.g., a larger square pixel array) of a regular shape that can be readily integrated in a semiconductor device. Further, ToF sensors **202** being oriented in perpendicular subsets can enable sharing of a given ToF sensor **202** by pixel sensors **206** of two image sensors **204**, as described below, thereby enabling an increased pixel area of an array of pixel arrays **200** having the first example layout.

(30) FIGS. 2B and 2C are cross-sectional diagrams at the lines labeled 1-1 and 2-2 in FIG. 2A.

(31) FIG. 2B is a cross-section along a length of a ToF sensor **202** (e.g., ToF sensor **202a**). As shown in FIG. 2B, the ToF sensor **202** includes a photodiode **212** (e.g., including one or more germanium doped regions) formed in a substrate **210** (e.g., a silicon substrate), a set of p-type portions **214** (e.g., one or more p+ portions), a set of n-type portions **216** (e.g., one or more n+ portions), and isolation structures **218**. As further shown in FIG. 2B, a layer **220** (e.g., including an oxide layer, a dielectric layer, or the like) with a set of contacts **222** (not shown in FIG. 2A) may be formed on the ToF sensor **202**.

(32) FIG. 2C is an example cross-section across a width of a ToF sensor **202** and across a pixel sensor **206** of the image sensor **204**. As shown in FIG. 2C, photodiodes **224** of the pixel sensor **206** are formed in the substrate **210**, the image sensor **204** further includes an n-type portion **226** (e.g., an n+ portion) and a gate **228**. As further shown in FIG. 2C, the layer **220** and contacts **222** may be formed over the pixel sensor **206**. As shown in FIG. 2C, the ToF sensor **202** may be separated from the image sensor **204** by an isolation region **208** in the substrate **210**.

(33) In some implementations, as described above, the ToF sensors **202** surround the image sensor **204** such that the image sensor **204** is centered in the ToF sensors **202**. In some such cases, to enable the group of ToF sensors **202** to surround image sensor **204**, ToF sensors **202** in the group of ToF sensors **202** may be oriented in the same direction (rather than in different directions, as described above). That is, in some implementations, each ToF sensor **202** in the group of ToF sensors **202** may be oriented parallel to each other ToF sensor **202** in the group of ToF sensors **202**.

(34) FIG. 2D is a top view of a second example layout of the pixel array **200** in which each ToF sensor **202** in the group of ToF sensors **202** is oriented parallel to each other ToF sensor **202** in the group of ToF sensors **202**. As shown by the second example layout in FIG. 2D, ToF sensors **202a** through **202d** are oriented with respective lengths along a single direction (e.g., a vertical direction in FIG. 2D). In some implementations, a pattern enabled by the ToF sensors **202** being oriented in the same direction enables multiple pixel arrays **200** (and/or multiple portions of pixel arrays **200**)

having the second example layout to be arranged adjacent to one another to form a larger pixel array with a first overall dimension that is different from a second overall dimension (e.g., such that a rectangular array with non-equal dimensions is formed). For example, the pattern enabled by the second example layout may be used to form a rectangular array in which a length is greater than a width, which may be useful when, for example, a rectangular area is available on a semiconductor device for integration of the pixel array. Further, ToF sensors **202** being oriented in parallel to one another enables sharing of a given ToF sensor **202** by pixel sensors **206** of at least two image sensors **204** (e.g., four pixel sensors **206** of four different image sensors **204**), as described below, thereby increasing pixel area of an array of pixel arrays **200** having the second example layout.

(35) As indicated above, FIGS. 2A-2D are provided as examples. Other examples may differ from what is described with regard to FIGS. 2A-2D. Further, the ToF sensor **202** or the image sensor **204** shown in FIGS. 2A-2D may include additional elements, fewer elements, differently arranged elements, or elements having different relative sizes than those shown in FIGS. 2A-2D.

(36) In some implementations, distance information determined by ToF sensors **202** and color information determined by pixel sensors **206** of an image sensor **204** may be used by an imaging system (e.g., by various components of the imaging system, such as one or more processors, transistors, memories, or other components) including the pixel array **200** to generate 3D ToF color information. For example, a particular ToF sensor **202** may be paired with a particular pixel sensor **206** of the image sensor **204** such that an output of the particular ToF sensor **202** is used in conjunction with an output of the particular pixel sensor **206**. That is, the output of the particular ToF sensor **202** (e.g., a distance as measured by the particular ToF sensor **202**) may be combined with or otherwise associated with the output of the particular pixel sensor **206** (e.g., an intensity of a particular color of visible light as measured by the particular pixel sensor **206**) to create 3D ToF color information (e.g., information that identifies a distance and an intensity of the particular color of visible light) corresponding to a location of the particular pixel sensor **206**. 3D ToF color information can be similarly generated for additional ToF sensor **202**/pixel sensor **206** pairings, and a 3D ToF color image can be generated from the 3D ToF color information determined across the array.

(37) FIGS. 3A-3C are diagrams of example pixel arrays comprising multiple pixel arrays **200** that enable pairing of ToF sensors **202** and pixel sensors **206**. FIG. 3A is a diagram of a top view of a portion of a pixel array **300** that includes multiple pixel arrays **200** having the first example layout shown in FIG. 2A. As shown in FIG. 3A, multiple pixel arrays **200** having the first example layout are arranged adjacent to one another to form the pixel array **300**. As illustrated by the black arrows in FIG. 3A, a given ToF sensor **202** is paired with a single pixel sensor **206** in the pixel array **300**. That is, in the pixel array **300**, an output of a given ToF sensor **202** is to be used in conjunction with an output of only one pixel sensor **206** (e.g., such that each ToF sensor **202** is used in conjunction with a different pixel sensor **206**). In some implementations, due to the pairing within the pixel array **300**, a quantity of ToF sensors **202** is greater than a quantity of image sensors **204** in the pixel array **300** (e.g., the quantity of ToF sensors **202** is four times greater than the quantity of image sensor **204** when each image sensor **204** includes four pixel sensors **206**). Put another way, the quantity of ToF sensors **202** in the pixel array **300** matches the quantity of pixel sensors **206** (e.g., when each image sensor **204** includes four pixel sensors **206**). In some implementations, the pairing of a given ToF sensor **202** with a single pixel sensor **206** (enabled by the pixel array **300**) improves performance of the pixel array **300** in terms of, for example, accuracy or contour sharpness in a 3D ToF image generated based on information collected by the pixel array **300** (e.g., as compared to a pixel array in which a given ToF sensor **202** is paired with multiple pixel sensors **206**).

(38) FIG. 3B is a diagram of a top view of a portion of a pixel array **310** that includes multiple pixel arrays **200** having the first example layout shown in FIG. 2A, where the pixel arrays **200** are arranged such that a ToF sensor **202** is included in two different pixel arrays **200**. That is, in the

pixel array **310**, multiple pixel arrays **200** having the first example layout can be arranged to be partially overlapping such that a ToF sensor **202** is included in two different pixel arrays **200**, as shown by the labeled pixel arrays **200** in FIG. 3B. As illustrated by the black arrows in FIG. 3B, a given ToF sensor **202** is paired with two pixel sensors **206** in the pixel array **310**. Here, the output of the given ToF sensor **202** is to be used in conjunction with an output of a pixel sensor **206** in a first image sensor **204** and in conjunction with an output of a pixel sensor **206** in a second image sensor **204**. Put another way, in the pixel array **310**, an output of a given ToF sensor **202** is to be used in conjunction with an output of two pixel sensors **206** from different image sensors **204**. In some implementations, the pairing within the pixel array **310** causes a quantity of ToF sensors **202** to be greater than a quantity of image sensors **204** in the pixel array **300** (e.g., the quantity of ToF sensors **202** is two times greater than the quantity of image sensors **204** when each image sensor **204** includes four pixel sensors **206**). Put another way, the quantity of ToF sensors **202** in the pixel array **310** may be approximately one-half the quantity of pixel sensors **206** (e.g., when each image sensor **204** includes four pixel sensors **206**). In some implementations, the pairing of a given ToF sensor **202** with two pixel sensors **206** (enabled by the pixel array **310**) increases a pixel area of the pixel array **310** (e.g., as compared to the pixel array **300**, which has twice as many ToF sensors **202** as the pixel array **310**). Here, the increased pixel area can provide improved performance of the pixel array **310** in terms of, for example, color saturation, color accuracy, noise, contrast, brightness, hue and saturation, or light sensitivity, without significantly impacting performance of the pixel array **310** in terms of, for example, accuracy or contour sharpness.

(39) FIG. 3C is a diagram of a top view of a portion of a pixel array **320** that includes multiple pixel arrays **200** having the second example layout shown in FIG. 2D, where the pixel arrays **200** are arranged such that a ToF sensor **202** is included in four different pixel arrays **200**. That is, in the pixel array **320**, multiple pixel arrays **200** having the second example layout can be arranged to be partially overlapping such that a ToF sensor **202** is included in four different pixel arrays **200**, as shown by the labeled pixel arrays **200** in FIG. 3C, where the ToF sensor **202** at the center of the four labeled pixel arrays **200** having the second example layout is included in all four of the labeled pixel arrays **200**. As illustrated by the black arrows in FIG. 3C, a given ToF sensor **202** is paired with four pixel sensors **206** in the pixel array **320**. Here, the output of the given ToF sensor **202** is to be used in conjunction with an output of a pixel sensor **206** in a first image sensor **204**, in conjunction with an output of a pixel sensor **206** in a second image sensor **204**, in conjunction with an output of a pixel sensor **206** in a third image sensor **204**, and in conjunction with an output of a pixel sensor **206** in a fourth image sensor **204**. Put another way, in the pixel array **320**, an output of a given ToF sensor **202** is to be used in conjunction with an output of four pixel sensors **206** from different image sensors **204**. In some implementations, the pairing within the pixel array **320** causes a quantity of ToF sensors **202** to match (e.g., be equal to) a quantity of image sensors **204** in the pixel array **320** (e.g., the quantity of ToF sensors **202** is equal to the quantity of image sensors **204** when each image sensor **204** includes four pixel sensors **206**). Put another way, the quantity of ToF sensors **202** in the pixel array **320** may be one quarter of the quantity of pixel sensors **206** (e.g., when each image sensor **204** includes four pixel sensors **206**). In some implementations, the pairing of a given ToF sensor **202** with four pixel sensors **206** (enabled by the pixel array **320**) increases a pixel area of the pixel array **320** (e.g., as compared to the pixel array **300** which has four times as many ToF sensors **202** as the pixel array **320**). Here, the increased pixel area can provide further improved performance of the pixel array **320** in terms of, for example, color saturation, color accuracy, noise, contrast, brightness, hue and saturation, or light sensitivity. Notably, accuracy or contour sharpness of a 3D ToF image may decreased by the sharing of a single ToF sensor **202** with four pixel sensors **206** (e.g., as compared to one-to-one pairing of ToF sensors **202** and pixel sensors **206**).

(40) As indicated above, FIGS. 3A-3C are provided as examples. Other examples may differ from what is described with regard to FIGS. 3A-3C.

(41) FIG. 4 is a diagram of an example pixel array **400** described herein. In some implementations, the example pixel array **400** illustrated in FIG. 4 may include, or may be included in, the pixel array **200**, the pixel array **300**, the pixel array **310**, the pixel array **320**, or a portion of any of the pixel arrays described herein. Further, the example pixel array **400** is shown for illustrative purpose, and the pixel array **400** can be adjusted to match the pixel array **200**, the pixel array **300**, the pixel array **310**, the pixel array **320**, or a portion of any of the pixel arrays described herein.

(42) As shown in FIG. 4, the pixel array **400** may include a set of ToF sensors **202** and a set of pixel sensors **206** of an image sensor **204**. In some implementations, the ToF sensors **202** and the pixel sensors **206** are configured in the example layout for the pixel array **200** shown in FIG. 2A or FIG. 2D.

(43) The ToF sensors **202** and the pixel sensors **206** may be formed in a substrate **210**, which may include a semiconductor die substrate, a semiconductor wafer, or another type of substrate in which semiconductor pixels may be formed. In some implementations, the substrate **210** is formed of silicon (Si), a material including silicon, a III-V compound semiconductor material such as gallium arsenide (GaAs), a silicon on insulator (SOI), or another type of semiconductor material that is capable of generating a charge from photons of incident light.

(44) Each ToF sensor **202** may include a photodiode **212**. A photodiode **212** may include a region of the substrate **210** that is doped with a plurality of types of ions to form a p-n junction or a PIN junction (e.g., a junction between a p-type portion, an intrinsic (or undoped) type portion, and an n-type portion) for use in ToF sensing. For example, the substrate **210** may be doped with an n-type dopant to form a first portion (e.g., an n-type portion) of a photodiode **212** and a p-type dopant to form a second portion (e.g., a p-type portion) of the photodiode **212**. In some implementations, the photodiode **212** includes one or more germanium doped regions. A photodiode **212** may be configured to absorb photons of incident light, such as infrared light, near-infrared light, light of approximately 1550 nanometers (nm), or the like. The absorption of photons causes a photodiode **212** to accumulate a charge (referred to as a photocurrent) due to the photoelectric effect. Here, photons bombard the photodiode **212**, which causes emission of electrons of the photodiode **212**. The emission of electrons causes the formation of electron-hole pairs, where the electrons migrate toward the cathode of the photodiode **212** and the holes migrate toward the anode, which produces the photocurrent.

(45) Each pixel sensor **206** may include a photodiode **224**. A photodiode **224** may include a region of the substrate **210** that is doped with a plurality of types of ions to form a p-n junction or a PIN junction (e.g., a junction between a p-type portion, an intrinsic (or undoped) type portion, and an n-type portion). For example, the substrate **210** may be doped with an n-type dopant to form a first portion (e.g., an n-type portion) of a photodiode **224** and a p-type dopant to form a second portion (e.g., a p-type portion) of the photodiode **224**. A photodiode **224** may be configured to absorb photons of incident light, such as visible light (e.g., red light, green light, blue light, or light having a wavelength of less than approximately 800 nm, among other examples). The absorption of photons causes a photodiode **224** to accumulate a charge (referred to as a photocurrent) due to the photoelectric effect. Here, photons bombard the photodiode **224**, which causes emission of electrons of the photodiode **224**. The emission of electrons causes the formation of electron-hole pairs, where the electrons migrate toward the cathode of the photodiode **224** and the holes migrate toward the anode, which produces the photocurrent.

(46) An isolation structure **218** (shown as including isolation structure **218a** and isolation structure **218b**) may be included in the substrate **210** between adjacent elements of the pixel array **400** (e.g., adjacent ToF sensors **202**, adjacent pixel sensors **206**, and/or between a ToF sensor **202** and a pixel sensor **206**). The isolation structure **218** may provide optical isolation by blocking or preventing diffusion or bleeding of light from one ToF sensor **202**/pixel sensor **206** to another ToF sensor **202**/pixel sensor **206**, thereby reducing crosstalk between adjacent elements of the pixel array **400**. The isolation structure **218** may include trenches or deep trench isolation (DTI) structures filled

with, for example, an oxide material. In some implementations, the isolation structure **218** may be formed in a grid layout in which the isolation structure **218** extends around the perimeters of the ToF sensors **202** and/or the pixel sensors **206** in the pixel array **400** and intersects at various locations of the pixel array **400**. In some implementations, a portion of the isolation structure **218** (e.g., isolation structure **212a**) is formed in the isolation region **208** of the substrate **210** or is formed adjacent to the isolation region **208**.

(47) The routing structure **401** is a structure associated with interconnecting the devices of the pixel array **400** (e.g., the ToF sensors **202** and the pixel sensors **206** of the image sensor **204**) with wiring (e.g., a metallization layer, not shown in FIG. 4). In some implementations, the routing structure **401** may include one or more dielectric layers, one or more metal layers, one or more contacts, one or more vias, or one or more passivation layers, among other examples. In some implementations, the routing structure **401** is formed using a back end of line (BEOL) process. Notably, particular details of the features of the routing structure **401** are not shown or described with specificity.

(48) The oxide layer **402** may function as a dielectric buffer layer between the photodiodes **212** and the photodiodes **224** and the layers above the photodiodes **212** and the photodiodes **224**. The oxide layer **402** may include an oxide material such as a silicon oxide (SiO.sub.x) (e.g., silicon dioxide (SiO.sub.2)), a hafnium oxide (HfO.sub.x), a tantalum oxide (TaO.sub.x), an aluminum oxide (AlO.sub.x), or another type of dielectric oxide material. In some implementations, another type of dielectric material is used in place of the oxide layer **402**, such as a silicon nitride (Si.sub.xN.sub.y), a silicon carbide (SiC.sub.x), a titanium nitride (TiN.sub.x), or a tantalum nitride (TaN.sub.x).

(49) A grid structure **404** may be included over and/or on the oxide layer **402**. The grid structure **404** may include a plurality of interconnected columns formed from a plurality of layers that are etched to form the columns. The grid structure **404** may surround the perimeters of the ToF sensors **202** and/or the pixel sensors **206** and may be configured to provide additional crosstalk reduction and/or mitigation in combination with the isolation structure **218**.

(50) In some implementations, the sidewalls of the grid structure **404** are substantially straight and parallel (e.g., the sidewalls are at an approximately 90 degree angle relative to a top surface of the grid structure **404**). In some implementations, the sidewalls of the grid structure **404** are angled or tapered. In these examples, the sidewalls may taper between the top and the bottom of the grid structure **404** at an angle (e.g., at a non-90-degree angle, such as at an angle that is greater than approximately 90 degrees and less than or equal to approximately 120 degrees) relative to the top surface of the grid structure **404** such that the bottom of the grid structure **404** is wider relative to the top of the grid structure **404**. For example, in some implementations, the sidewalls may taper between the top and the bottom of the grid structure **404** at a non-90-degree angle, such as at an angle that is greater than approximately 90 degrees and less than or equal to approximately 120 degrees, relative to the top surface of the grid structure **404**. However, other angles of the sidewalls relative to the top surface of the grid structure **404** are within the scope of the present disclosure. In some implementations, the particular angle of the sidewalls may be based on an amount of incident light that the grid structure **404** is to block (e.g., a greater angle may block a lesser amount of light relative to a smaller angle). The grid structure **404** may include a plurality of layers over and/or on the oxide layer **402**. The grid structure **404** may include one or more metal layers (or metal-containing layers) and one or more dielectric layers, and may be referred to a composite metal grid (CMG).

(51) Respective color filter regions **406** may be included in the areas between the grid structure **404**. For example, color filter regions **406c** may be formed in between columns of the grid structure **404** over photodiodes **224** of the pixel sensors **206**, and color filter regions **406t** may be formed in between columns of the grid structure **404** over the ToF sensors **202**. Alternatively, the areas between the grid structure **404** may be completely filled with a passivation layer, and a color filter layer including the color filter regions **406** may be formed above the grid structure **404** on the

passivation layer.

(52) Each color filter region **406** may be configured to filter incident light to allow a particular wavelength of the incident light (or all wavelengths of incident light) to pass. For example, the color filter region **406c** included in the left pixel sensor **206** of the pixel array **400** may filter red light for the left pixel sensor **206** (and thus, the left pixel sensor **206** may be a red pixel sensor) and the color filter region **406c** included in the right pixel sensor **206** may filter green light for the right pixel sensor **206** (and thus, the right pixel sensor **206** may be a green pixel sensor), and so on. Here, the color filter regions **406t** included in the ToF sensors **202** of the pixel array **400** may be non-discriminating or non-filtering, meaning that all wavelengths of light are allowed to pass through the color filter region **406t** included in the ToF sensors **202** (e.g., for purposes of performing ToF sensing).

(53) A micro-lens layer **408** may be included above and/or on the color filter regions **406**. The micro-lens layer **408** may include a respective micro-lens for each of the pixel sensors **206**. For example, a micro-lens may be formed to focus incident light toward a photodiode **212** of a given ToF sensor **202**, while another micro-lens may be formed to focus incident light toward a photodiode **224** of a given pixel sensor **206**, and so on.

(54) As indicated above, FIG. 4 is provided as an example. Other examples may differ from what is described with regard to FIG. 4.

(55) FIGS. 5A-5L are diagrams of an example implementation described herein. The example implementation may be an example process or method for forming the pixel array **400**. In some implementations, the various example techniques and procedures described in connection with FIGS. 5A-5L may be used in connection with other pixel arrays described herein, such as the pixel array **200** described in connection with FIG. 2A, the pixel array **200** described in connection with FIG. 2D, the pixel array **300** described in connection with FIG. 3A, the pixel array **310** described in connection with FIG. 3B, and/or the pixel array **320** described in connection with FIG. 3C.

(56) As shown in FIG. 5A, the ToF sensors **202** and the pixel sensors **206** may be formed in the substrate **210**. The substrate **210** may include a silicon substrate, a substrate formed of a material including silicon, a III-V compound semiconductor substrate such as gallium arsenide (GaAs) substrate, a silicon on insulator (SOI) substrate, or another type of substrate is capable of generating a charge from photons of incident light.

(57) As shown in FIG. 5B, one or more semiconductor processing tools may form a plurality of photodiodes **212** and a plurality of photodiodes **224** in the substrate **210**. For example, the ion implantation tool **112** may dope the portions of the substrate **210** using an ion implantation technique to form respective photodiodes **212** for a plurality of ToF sensors **202** and photodiodes **224** for a plurality of pixel sensors **206**. The substrate **210** may be doped with a plurality of types of ions to form each photodiode **212** and each photodiode **224**.

(58) As shown in FIG. 5C, openings **218ao** may be formed in the substrate **210** in association with forming an isolation structure **218a** (e.g., a shallow trench isolation (STI) structure) in the substrate **210**. In particular, the openings **218ao** may be formed such that the isolation structure **218a** may be formed between photodiodes **212**, between photodiodes **224**, and between photodiodes **212** and photodiodes **224** of the pixel array **400**. In some implementations, one or more semiconductor processing tools may be used to form the one or more openings **218ao** in the substrate **210**. For example, the deposition tool **102** may form a photoresist layer on the substrate **210**, the exposure tool **104** may expose the photoresist layer to a radiation source to pattern the photoresist layer, the developer tool **106** may develop and remove portions of the photoresist layer to expose the pattern, and the etch tool **108** may etch portions of substrate **210** to form the openings **218ao** in the substrate **210**. In some implementations, a photoresist removal tool removes the remaining portions of the photoresist layer (e.g., using a chemical stripper and/or another technique) after the etch tool **108** etches the substrate **210**.

(59) As shown in FIG. 5D, the openings **218ao** may be filled with an oxide material to form

isolation structure **218a**. In particular, a semiconductor processing tool (e.g., the deposition tool **102**) may deposit an oxide material (e.g., a silicon oxide ($\text{SiO}_{\text{sub.x}}$), a tantalum oxide ($\text{Ta}_{\text{sub.x}}\text{O}_{\text{sub.y}}$), or another type of oxide) such that the oxide material is formed in and on the openings **218ao**. The semiconductor processing tool may deposit the oxide material using various PVD techniques, CVD techniques and/or ALD techniques, such as sputtering, PECVD, HDP-CVD, SACVD, or PEALD.

(60) As shown in FIG. 5E, a routing structure **401** may be formed on the pixel array **400** (e.g., on the ToF sensors **202** and the pixel sensors **206**) and the isolation structure **218a**. For example, one or more of semiconductor processing tools (e.g., the deposition tool **102**, the exposure tool **104**, the developer tool **106**, the etch tool **108**, the planarization tool **110**, or the like) may perform a BEOL process that forms the routing structure **401** on the pixel array **400** and the isolation structure **218a**.

(61) As shown in FIG. 5F, the pixel array **400** may be flipped for further processing. In some implementations, the pixel array **400** may be affixed to a carrier wafer (e.g., using an adhesive such as a tape or a glue, among other examples) prior to being flipped. In some implementations, after the pixel array **400** is affixed to the carrier wafer and is flipped, the substrate **210** may be thinned in order to remove a portion of the substrate **210** from the back side of the pixel array **400**. For example, the planarization tool **110** may perform a wafer thinning process to remove a portion of the substrate **210** from the back side of the pixel array **400**. In some implementations, the planarization tool **110** may grind the substrate **210** to thin the substrate **210** on the back side of the pixel array **400** after the pixel array **400** is affixed to the carrier wafer.

(62) As shown in FIG. 5G, openings **218bo** may be formed in the substrate **210** in association with forming an isolation structure **218b** (e.g., a DTI structure) in the substrate **210**. In particular, the openings **218bo** may be formed such that (at least some portion of) the isolation structure **218b** may be formed between photodiodes **212**, between photodiodes **224**, and between photodiodes **212** and photodiodes **224** of the pixel array **400**. In some implementations, one or more semiconductor processing tools may be used to form the one or more openings **218bo** in the substrate **210**. For example, the deposition tool **102** may form a photoresist layer on the substrate **210**, the exposure tool **104** may expose the photoresist layer to a radiation source to pattern the photoresist layer, the developer tool **106** may develop and remove portions of the photoresist layer to expose the pattern, and the etch tool **108** may etch portions of substrate **210** to form the openings **218bo** in the substrate **210**. In some implementations, a photoresist removal tool removes the remaining portions of the photoresist layer (e.g., using a chemical stripper and/or another technique) after the etch tool **108** etches the substrate **210**.

(63) As shown in FIG. 5H, the openings **218bo** may be filled with an oxide material to form isolation structure **218b** and the oxide layer **402**. In particular, a semiconductor processing tool (e.g., the deposition tool **102**) may deposit the oxide layer **402** (e.g., a silicon oxide ($\text{SiO}_{\text{sub.x}}$), a tantalum oxide ($\text{Ta}_{\text{sub.x}}\text{O}_{\text{sub.y}}$), or another type of oxide) such that the oxide layer **402** is formed in and on the isolation structure **218b**. The semiconductor processing tool may deposit the oxide layer **402** using various PVD techniques, CVD techniques and/or ALD techniques, such as sputtering, PECVD, HDP-CVD, SACVD, or PEALD.

(64) As shown in FIG. 5I, one or more layers may be formed over and/or on the oxide layer **402** as part of forming the grid structure **404**. For example, the deposition tool **102** may deposit the one or more layers (e.g., one or more metal layers and/or one or more dielectric layers) to form the layer from which the grid structure **404** will be formed. The deposition tool **102** may deposit the one or more layers using various PVD techniques, CVD techniques and/or ALD techniques, such as sputtering, PECVD, HDP-CVD, SACVD, or PEALD.

(65) As shown in FIG. 5J, openings **404o** may be formed in the plurality of layers to form the grid structure **404**. The openings **404o** may be formed over the photodiodes **212** of the ToF sensors **202** and the photodiodes **224** of the pixel sensors **206** such that color filter regions may be filled in the openings **404o**. In some implementations, the openings **404o** and the grid structure **404** are formed

using a photoresist (e.g., deposited by the deposition tool **102**), where a pattern in the photoresist is formed by exposing the photoresist to a radiation source (e.g., using the exposure tool **104**) and removing either the exposed portions or the non-exposed portions of the photoresist (e.g., using developer tool **106**). Here, the etch tool **108** etches the openings **404o** through the one or more layers associated with the grid structure **404** based on the patterned photoresist.

(66) As shown in FIG. 5K, respective color filter regions **406** may be formed for each of the ToF sensors **202** and pixel sensors **206** in the pixel array **400** in the openings **404o**. Each color filter region **406** may be formed in between the grid structure **404** to reduce color mixing between adjacent pixel sensors **206**. A semiconductor processing tool (e.g., the deposition tool **102**) may deposit the color filter regions **406** using various PVD techniques, CVD techniques and/or ALD techniques, such as sputtering, PECVD, HDP-CVD, SACVD, or PEALD.

(67) As shown in FIG. 5L, the micro-lens layer **408** including a plurality of micro-lenses is formed over and/or on the color filter regions **406**. The micro-lens layer **408** may include a respective micro-lens for each of the ToF sensors **202** and each of the pixel sensors **206** included in the pixel array **400**.

(68) As indicated above, FIGS. 5A-5L are provided as an example. Other examples may differ from what is described with regard to FIGS. 5A-5L.

(69) In some implementations, one or more pixel sensors **206** in a group of pixel sensors **206** and/or one or more ToF sensors **202** in a group of ToF sensors **202** of a pixel array **200** may include respective high absorption regions. FIGS. 6A and 6B are diagrams of example pixel arrays in which pixel sensors **206** and ToF sensors **202** include respective high absorption regions. FIG. 6A is a diagram of a top view of an example pixel array **600** having the first example layout in which pixel sensors **206** and ToF sensors **202** include respective high absorption regions **602**. FIG. 6B is an example cross-sectional view of a pixel array **400** in which pixel sensors **206** and ToF sensors **202** include respective high absorption regions **602**.

(70) A high absorption region **602** may increase the absorption of incident light for a ToF sensor **202**/pixel sensor **206** (thereby increasing the quantum efficiency (QE) of the ToF sensor **202**/pixel sensor **206**) by modifying or changing the orientation of the refractive interface between the oxide layer **402** of the photodiode **212**/photodiode **224** and the oxide layer **402**. A high absorption region **602** may include a structure having angled walls such that the structure is approximately triangular shaped. The angled walls of the high absorption region **602** may be angled relative to the top surface of the substrate **210**, and may be angled inward toward a photodiode **212**/photodiode **224**. Thus, the angled walls change the orientation of the interface by causing the interface to be diagonal relative to the orientation of the top surface of the substrate **210**. This change in orientation may result in a change in refraction relative to the flat surface of the top layer of the substrate **210** for the same angle of incidence of incident light. For example, increasing the inward angle of the angled walls relative to the top surface of the substrate **210** may decrease reflections of incident light traveling at a high angle of incidence relative to the top surface of the substrate **210** because the increased inward angle decreases the angle of incidence of the incident light relative to the angled walls of the high absorption region **602**. As a result, the high absorption region **602** is capable of directing wider angles of incident light toward the center of the photodiode **212**/photodiode **224** than if no high absorption region **602** were included in the pixel sensor **206**. In some implementations, the high absorption region **602** can improve QE, increase intensity of light received by a given sensor of pixel array **200** (e.g., by increasing an amount of light directed to the photodiode **212**/photodiode **224**), and increase a pixel area of a given sensor of the pixel array **200**, thereby improving resolution achieved by the sensors of the pixel array **200**.

(71) In some implementations, all of the ToF sensors **202** and pixel sensors **206** of the pixel array **200** include a high absorption region **602**. In some implementations, a subset of the ToF sensors **202** and the pixel sensors **206** of the pixel array **200** include a high absorption region **602**. In some implementations, all of a particular type of pixel sensor **206** (e.g., a red pixel sensor, a green pixel

sensor, a blue pixel sensor, a ToF sensor **202**, among other examples) of the pixel array **200** include a high absorption region **602**. In some implementations, a first subset of a particular type of pixel sensor **206** of the pixel array **200** includes a high absorption region **602**, and a high absorption region **602** is omitted from a second subset of the particular type of pixel sensor **206** of the pixel array **200**. High absorption regions **602** may be included in or excluded from the pixel sensors **206** of the pixel array **200** based on various factors, such as a target quantum efficiency for the pixel sensors **206**, the intended application or use case for the pixel array **200**, or the like. For example, high absorption regions **602** might be included in the pixel sensors **206** of the pixel array **200** to achieve a high target quantum efficiency, or may be excluded from the pixel sensors **206** if a lower target quantum efficiency is specified. As another example, high absorption regions **602** might be included in the pixel sensors **206** of the pixel array **200** if the intended application or use case for the pixel array **200** involves a large amount of expected off-angle or wide-angle incident light, or may be excluded from the pixel sensors **206** if the intended application or use case for the pixel array **200** primarily involves coherent or narrow-angle incident light (e.g., such as light emitted from a laser).

(72) As shown in FIG. **6B**, a high absorption region **602** may be formed in the substrate **210** and may extend into a respective photodiode **212** of a ToF sensor **202** or a respective photodiode **224** of a pixel sensor **206**. For example, the substrate **210** may be patterned with a photoresist, and the photodiode region of a pixel sensor **206** may be etched based on the pattern in the photoresist to form a high absorption region **602** for the pixel sensor **206**. The high absorption region **602** may be filled with the oxide material of the oxide layer **402** during deposition of the oxide layer **402**.

(73) In some implementations, the high absorption regions **602** for a plurality of ToF sensors **202** and/or pixel sensors **206** may be the same size. For example, the high absorption regions **602** for a plurality of ToF sensor **202** and/or pixel sensors **206** may have the same height H . In some implementations, the high absorption regions **602** for the plurality of ToF sensor **202** and/or pixel sensors **206** may be different sizes. For example, the high absorption regions **602** for one or more ToF sensors **202** may have a different height H than the high absorption regions **602** for one or more pixel sensors **206**. In some implementations, the height H for the high absorption regions **602** for ToF sensors **202**, or for pixel sensors **206** of a particular type, may be the same height. In some implementations, the height H for the high absorption regions **602** for ToF sensors **202**, or for pixel sensors **206** of a particular type, may be different.

(74) In some implementations, the height H for the high absorption regions **602** for a first type of sensor in the pixel array **200** (e.g., ToF sensor **202**, a particular type of pixel sensor **206**) may be greater relative to the height H for high absorption regions **602** for a second type of sensor in the pixel array **200**. In some implementations, the height H for the high absorption region **602** may be based on a wavelength of incident light that is to be sensed or absorbed by the pixel sensor **206**. For example, the height H for the high absorption regions **602** included in the pixel array **200** may be increased as the wavelength of incident light that is to be sensed or absorbed increases, and may be decreased as the wavelength of incident light that is to be sensed or absorbed decreases. This is because increasing the height of a high absorption region **602** results in larger (longer) angled walls that can better accommodate longer wavelengths of light, whereas decreasing the height of a high absorption region **602** provides relatively smaller (shorter) angled walls for shorter wavelengths of light.

(75) As an example, the height H for the high absorption regions **602** included in blue pixel sensors of the pixel array **200** may be the smallest height (e.g., because blue light is the shortest wavelength). The height H for the high absorption regions **602** included in green pixel sensors of the pixel array **200** may be larger than the height H for the high absorption regions **602** included in the blue pixel sensors (e.g., because the wavelength of green light is greater than the wavelength of blue light). The height H for the high absorption regions **602** included in red pixel sensors of the pixel array **200** may be larger than the height H for the high absorption regions **602** included in the

green pixel sensors and the blue pixel sensors (e.g., because the wavelength of red light is greater than the wavelength of green light, and the wavelength of blue light). The height H for the high absorption regions **602** included in ToF sensors **202** of the pixel array **200** may be larger than the height H for the high absorption regions **602** included in the red pixel sensors, the green pixel sensors, and the blue pixel sensors (e.g., because wavelengths of light to be received by the ToF sensor **202** is greater than the wavelength of red light, the wavelength of green light, and the wavelength of blue light).

(76) In some implementations, the width W of a high absorption region **602** may scale proportionally with the height H of the high absorption region **602** to ensure that the angle of the angled walls of the high absorption region **602** stays constant. In these examples, the width W of a high absorption region **602** may increase proportionally with an increase in height H of the high absorption region **602**, or may decrease proportionally with a decrease in height H to ensure that the angle of the angled walls of the high absorption region **602** stays constant.

(77) As indicated above, FIGS. **6A** and **6B** are provided as examples. Other examples may differ from what is described with regard to FIGS. **6A** and **6B**.

(78) FIG. **7** is a diagram illustrating absorption data associated with materials in which ToF sensors **202** and pixel sensors **206** described herein can be formed. The lines labeled “1” and “2” in FIG. **7** represent absorption rates of silicon and a silicon-based material used in a CMOS image sensor, respectively. The line labeled “3” in FIG. **7** represents an absorption rate of SiGe. As shown in FIG. **7**, the absorption rates of silicon and the silicon-based material used in the CMOS sensor decrease more rapidly with increasing wavelength as compared to the absorption rate of SiGe. Notably, the absorption rate of SiGe remains relatively high up to approximately 1550 nm, after which the absorption rate decreases. Due to the characteristics of SiGe, in some implementations, ToF sensors **202** can utilize a light source with a wavelength of approximately 1550 nm or higher. As such, SiGe is suitable for ToF sensing in low light conditions (e.g., at night) because SiGe absorbs invisible light (e.g., light having a wavelength greater than approximately 740 nm) better than silicon and a silicon-based material.

(79) As indicated above, FIG. **7** is provided as an example. Other examples may differ from what is described with regard to FIG. **7**.

(80) FIG. **8** is a diagram of example components of a device **800**. In some implementations, one or more of the semiconductor processing tools **102-112** and/or the wafer/die transport tool **114** may include one or more devices **800** and/or one or more components of device **800**. As shown in FIG. **8**, device **800** may include a bus **810**, a processor **820**, a memory **830**, a storage component **840**, an input component **850**, an output component **860**, and a communication component **870**.

(81) Bus **810** includes a component that enables wired and/or wireless communication among the components of device **800**. Processor **820** includes a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of processing component. Processor **820** is implemented in hardware, firmware, or a combination of hardware and software. In some implementations, processor **820** includes one or more processors capable of being programmed to perform a function. Memory **830** includes a random access memory, a read only memory, and/or another type of memory (e.g., a flash memory, a magnetic memory, and/or an optical memory).

(82) Storage component **840** stores information and/or software related to the operation of device **800**. For example, storage component **840** may include a hard disk drive, a magnetic disk drive, an optical disk drive, a solid state disk drive, a compact disc, a digital versatile disc, and/or another type of non-transitory computer-readable medium. Input component **850** enables device **800** to receive input, such as user input and/or sensed inputs. For example, input component **850** may include a touch screen, a keyboard, a keypad, a mouse, a button, a microphone, a switch, a sensor, a global positioning system component, an accelerometer, a gyroscope, and/or an actuator. Output

component **860** enables device **800** to provide output, such as via a display, a speaker, and/or one or more light-emitting diodes. Communication component **870** enables device **800** to communicate with other devices, such as via a wired connection and/or a wireless connection. For example, communication component **870** may include a receiver, a transmitter, a transceiver, a modem, a network interface card, and/or an antenna.

(83) Device **800** may perform one or more processes described herein. For example, a non-transitory computer-readable medium (e.g., memory **830** and/or storage component **840**) may store a set of instructions (e.g., one or more instructions, code, software code, and/or program code) for execution by processor **820**. Processor **820** may execute the set of instructions to perform one or more processes described herein. In some implementations, execution of the set of instructions, by one or more processors **820**, causes the one or more processors **820** and/or the device **800** to perform one or more processes described herein. In some implementations, hardwired circuitry may be used instead of or in combination with the instructions to perform one or more processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(84) The number and arrangement of components shown in FIG. **8** are provided as an example. Device **800** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **8**. Additionally, or alternatively, a set of components (e.g., one or more components) of device **800** may perform one or more functions described as being performed by another set of components of device **800**.

(85) FIG. **9** is a flowchart of an example process **900** associated with forming a pixel array. In some implementations, one or more process blocks of FIG. **9** may be performed by one or more semiconductor processing tools (e.g., one or more of the semiconductor processing tools **102-112**). Additionally, or alternatively, one or more process blocks of FIG. **9** may be performed by one or more components of device **800**, such as processor **820**, memory **830**, storage component **840**, input component **850**, output component **860**, and/or communication component **870**.

(86) As shown in FIG. **9**, process **900** may include forming a group of ToF sensors of a pixel array (block **910**). For example, the one or more semiconductor processing tools **102-112** may form a group of ToF sensors (e.g., a group of ToF sensors **202**) of a pixel array (e.g., a pixel array **200**, **300**, **310**, or **320**), as described above.

(87) As further shown in FIG. **9**, process **900** may include forming an image sensor of the pixel array, the image sensor including a group of pixel sensors, wherein a ToF sensor of the group of ToF sensors is paired with a pixel sensor in the group of pixel sensors of the image sensor (block **920**). For example, the one or more semiconductor processing tools **102-112** may form an image sensor (e.g., image sensor **204**) of the pixel array, the image sensor including a group of pixel sensors (e.g., a group of pixel sensors **206**), as described above. In some implementations, a ToF sensor of the group of ToF sensors is paired with a pixel sensor in the group of pixel sensors of the image sensor.

(88) Process **900** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

(89) In a first implementation, process **900** includes forming respective high absorption regions (e.g., respective high absorption regions **602**) in at least one pixel sensor of the group of pixel sensors and at least one ToF sensor of the group of ToF sensors.

(90) In a second implementation, alone or in combination with the first implementation, the image sensor is surrounded by the group of ToF sensors such that at least one ToF sensor of the group of ToF sensors is between the image sensor and another image sensor (e.g., another image sensor **204**) of the pixel array.

(91) In a third implementation, alone or in combination with one or more of the first and second implementations, the ToF sensor is paired with another pixel sensor (e.g., another pixel sensor **206**),

the other pixel sensor being included in a group of pixel sensors of another image sensor (e.g., another image sensor **204**).

(92) Although FIG. **9** shows example blocks of process **900**, in some implementations, process **900** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **9**. Additionally, or alternatively, two or more of the blocks of process **900** may be performed in parallel.

(93) In this way, a pixel array may include a plurality of ToF sensors and a plurality of image sensors (e.g., a plurality of CMOS image sensors, such as a plurality of RGB image sensors). Here, outputs of the plurality of ToF sensors and the plurality of image sensors may be used to generate a 3D ToF color image (rather than a distance image or a color image), which may be useful in, for example, 3D module construction, and which can be used in a variety of applications, such as a medical application, a VR application, an AR application, a 3D printing application, or an autonomous vehicle application, among other examples.

(94) As described in greater detail above, some implementations described herein provide a pixel array. The pixel array includes a group of ToF sensors. The pixel array includes an image sensor comprising a group of pixel sensors, where the image sensor is arranged among the group of ToF sensors such that the image sensor is adjacent to each ToF sensor in the group of ToF sensors.

(95) As described in greater detail above, some implementations described herein provide a pixel array. The pixel array includes a plurality of ToF sensors. The pixel array includes a plurality of image sensors, each of the plurality of image sensors comprising a respective group of pixel sensors, where an image sensor of the plurality of image sensors is arranged between a subset of ToF sensors of the plurality of ToF sensors, and where each pixel sensor in a group of pixel sensors of the image sensor and each ToF sensor in the subset of ToF sensors include respective high absorption regions.

(96) As described in greater detail above, some implementations described herein provide a method. The method includes forming a group of ToF sensors of a pixel array. The method includes forming an image sensor of the pixel array, the image sensor including a group of pixel sensors, where a ToF sensor of the group of ToF sensors is paired with a pixel sensor in the group of pixel sensors of the image sensor.

(97) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: forming a plurality of time-of-flight (ToF) sensors of a pixel array; and forming an image sensor, of the pixel array, including a plurality of pixel sensors, wherein each ToF sensor of the plurality of ToF sensors is electrically connected to a different pixel sensor of the plurality of pixel sensors, and wherein, in a top view of the pixel array: each of the plurality of ToF sensors is spaced away from the image sensor, and a top surface of each of the plurality of ToF sensors is coplanar with a top surface of two pixel sensors, of the plurality of pixel sensors, wherein the two pixel sensors are in contact with each other.
2. The method of claim 1, further comprising forming respective high absorption regions in at least one pixel sensor of the plurality of pixel sensors and at least one ToF sensor of the plurality of ToF sensors.

3. The method of claim 1, wherein the image sensor is surrounded by the plurality of ToF sensors such that at least one ToF sensor of the plurality of ToF sensors is between the image sensor and another image sensor of the pixel array.
4. The method of claim 1, wherein at least one ToF sensor of the plurality of ToF sensors is electrically connected to a pixel sensor of a group of pixel sensors of another image sensor.
5. A method, comprising: forming a plurality of time-of-flight (ToF) sensors, and a plurality of pixel sensors of an image sensor, in a substrate, wherein each of the plurality of ToF sensors is spaced away from the image sensor, and wherein a top surface of each of the plurality of ToF sensors is coplanar with a top surface of two pixel sensors of the plurality of pixel sensors, wherein the two pixel sensors are in contact with each other; and forming an oxide layer over the substrate and a plurality of absorption regions, different from the image sensor, extending from the oxide layer into a portion of the substrate, wherein the plurality of absorption regions are for the plurality of ToF sensors and the image sensor, and wherein the plurality of absorption regions increases an absorption of incident light for the plurality of ToF sensors and the image sensor.
6. The method of claim 5 wherein each of the oxide layer and the plurality of absorption regions comprises an oxide material.
7. The method of claim 5, wherein the plurality of absorption regions extend from a bottom surface of the oxide layer and into a top surface of the substrate.
8. The method of claim 5, wherein each of the plurality of absorption regions extends into a respective photodiode for each of the plurality of ToF sensors and the image sensor.
9. The method of claim 5, wherein each of the plurality of absorption regions comprises a structure having angled walls such that the structure is triangular shaped.
10. The method of claim 5, wherein each of the plurality of absorption regions has a same height.
11. The method of claim 5, wherein the plurality of absorption regions includes a first absorption region having a first height and a second absorption region having a second height different from the first height.
12. A method, comprising: forming a group of time-of-flight (ToF) sensors, and a plurality of pixel sensors, in a substrate; forming a plurality of photodiodes corresponding to the group of ToF sensors, wherein each of the group of ToF sensors comprises at least two photodiodes, of the plurality of photodiodes, spaced away from the plurality of pixel sensors, and wherein a top surface of each of the plurality of ToF sensors is coplanar with a top surface of two pixel sensors of the plurality of pixel sensors wherein the two pixel sensors are in contact with each other; forming an oxide layer over the group of ToF sensors, the plurality of pixel sensors, and the plurality of photodiodes, wherein the oxide layer is a distance away from, and does not intersect with, the group of ToF sensors, the plurality of pixel sensors, and the plurality of photodiodes; and forming a plurality of color filter regions, corresponding to the group of ToF sensors and the pixel sensor, over the oxide layer.
13. The method of claim 12, further comprising: forming an isolation structure in the substrate.
14. The method of claim 13, further comprising: forming openings in the substrate in association with forming the isolation structure.
15. The method of claim 13, wherein the isolation structure is formed between the plurality of photodiodes.
16. The method of claim 13, wherein the isolation structure comprises an oxide material.
17. The method of claim 12, further comprising: forming a grid structure over the oxide layer, wherein the plurality of color filter regions are formed over the grid structure.
18. The method of claim 17, further comprising: forming one or more dielectric or metal layers over the oxide layer, wherein the grid structure is formed in the one or more dielectric or metal layers.
19. The method of claim 18, wherein forming the grid structure comprises: forming a plurality of openings in the one or more dielectric or metal layers, wherein the plurality of color filter regions

are formed in the plurality of openings.

20. The method of claim 12, further comprising: forming a micro-lens layer over the plurality of color filter regions.
